

Monolith - DMS Replication

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Replication Summary

Replication Stats

Tables	Replication Groups	Total Elapsed Time (hrs)	Average Throughput
299	9	1063	41K

Replication Groups

Table Name	Jan'23 Size	DMS Tasks
PSP_LEDGER_BALANCE	188	Task-1
PSP_MONEY_MOVEMENT_TRANSACTION	354	
PSP_ENTRY_DETAIL_RECORD	1123	
PSP_FINANCIAL_TRANS_STATE	1114	
PSP_FINANCIAL_TRANSACTION	672	
PSP_PAYCHECK	569	
PSP_PAYCHECK_SPLIT	335	Task-1
PSP_COMPANY_EVENT	393	Task-2a
PSP_COMPANY_EVENT_DETAIL	1004	Task-2b
PSP_COMPANY_EVENT_EMAIL_PARAM	758	Task-2c
PSP_COMPENSATION	343	Taks-3
PSP_PROPERTY_AUDIT	372	
PSP_QBDT_PAYCHECK_INFO	456	
PSP_QBDT_PAYLINE_INFO	628	
PSP_PSTUB_PAY_ITEM	1198	Taks-4
PSP_TAX	1367	Task-5
PSP_ENTITLEMENT_MESSAGE	295	Task-6
PSP_PAYCHECK_USAGE	221	
PSP_DISBURSE_ADVICE_TAX_LIAB	183	
PSP_QBDT_TRANSACTION_INFO	176	
PSP_PAYSTUB	171	
PSP_DEDUCTION	170	
PSP_PSTUB_PAID_TIMEOFF_ITEM	139	
PSP_PSTUB_EMPLOYEE_INFO	125	
Rest of all tables	?	Task-7

Source - [PSP - Monolith Database Partitioning](#)

Note: Replication Groups are still being experimented but finalised yet

Replication Details

[PSP - Monolith - DMS Replication](#)

Replication Task Configuration

Instance class	Engine version	Storage	Multi AZ	Comments
dms.r5.16xlarge	3.4.7	500	No	Recommended

Fine Tuning Parameters

Full Load

Task Settings Parameters	Task Settings Values	Finalized	Explored	Comments
Target table preparation mode	Truncate	No	No	Helps only in Full Load. No significance in CDC

Maximum number of tables to load in parallel	16	No	Yes	16 is working well. May need to change per migration task group
Commit rate during full load	50000	Yes	Yes	Working without any memory issues

Change Data Capture

Task Settings Parameters	Task Settings Values	Finalized	Explored	Comments
Batch-optimized apply	No	No	No	There is no requirement to keep referential integrity on the target (foreign keys aren't enabled)

Recommendation/Best Practices

Populating a Database

- Follow the best practices like [Remove Indexes](#), [Remove Foreign Key Constraints](#) and other best practices during data population

Enable all control tables for debugging

Control table settings

History timeslot (minutes) [Info](#)

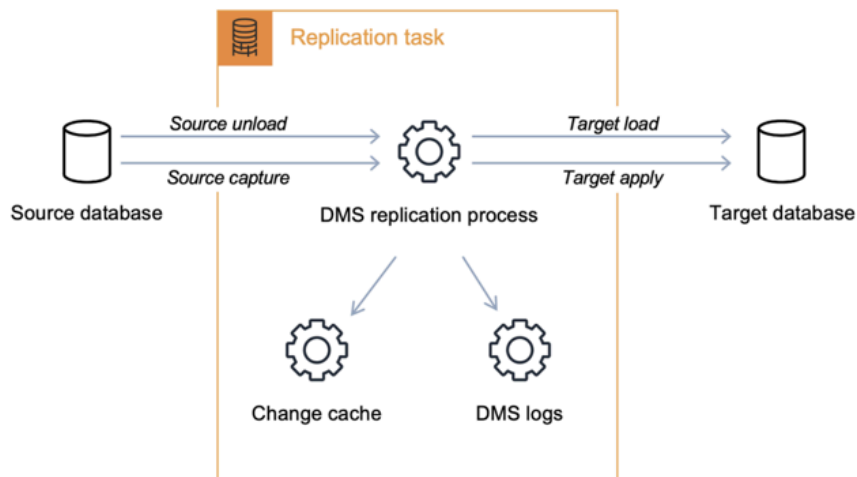
Create control table in target using schema [Info](#)

Enable control table	Name in target	Enable
Apply exceptions	awsdms_apply_exceptions	☐
Replication status	awsdms_status	☒
Suspended tables	awsdms_suspended_tables	☒
Replication history	awsdms_history	☒

DMS Internals

Replication Task

Conceptually, an [AWS DMS replication task](#) performs two distinct functions



Full Load

Data is extracted from the source in a bulk extract manner and loaded directly into the target

Ongoing replication, or change data capture (CDC)

You can also use an AWS DMS task to capture ongoing changes to the source data store while you are migrating your data to a target. The change capture process that AWS DMS uses when replicating ongoing changes from a source endpoint collects changes to the database logs by using the database engine's native API.

In the CDC process, the replication task is designed to stream changes from the source to the target, using in-memory buffers to hold data in-transit. If the in-memory buffers become exhausted for any reason, the replication task will spill pending changes to the Change Cache on disk

In AWS DMS, there are two methods for reading the redo logs when doing [change data capture \(CDC\) for Oracle as a source](#): Oracle LogMiner and AWS DMS Binary Reader.

- **LogMiner** is an Oracle API to read the online redo logs and archived redo log files.
- **Binary Reader** is an AWS DMS method that reads and parses the raw redo log files directly

Postgres Target

COPY

COPY command is used on the Postgres target to maximize the throughput. COPY command is faster due to

- Network latency and round-trip delays
 - There is an additional round trip latency for each INSERT which is avoided as the entire data is sent over the network and applied on the target
- Per-statement overheads in PostgreSQL
 - Typical [path of a query](#) includes [Parser](#) → [Analyser](#) → [Rewriter](#) → [Planner](#) → [Executor](#). But during COPY, [only Executor is involved](#) and avoid overhead of repeatedly parsing and planning.
- Context switches and scheduler delays

- Context switching and schedule delays are completely avoided as the postgres doesn't need to wait for any rows
- COPY-specific optimisations for bulk loading
 - Executor can also be bypassed if there is no generated key and any default values are constants (pre-calculate).
 - Combining [COPY with CREATE TABLE or TRUNCATE](#) can do even more tricks for making the load faster by bypassing the normal transaction book-keeping needed in a multi-client database. Not applicable if the logical replication is required.
- COMMIT costs, if for people doing one commit per insert
 - Large batch commit

References

- [How does COPY work and why is it so much faster than INSERT?](#)
- [Difference between INSERT and COPY](#)

Pipeline Mode

[Pipeline mode](#) allows applications to send a query without having to read the result of the previously sent query. Taking advantage of the pipeline mode, a client will wait less for the server, since multiple queries/results can be sent/received in a single network transaction.

While pipeline mode provides a significant performance boost, writing clients using the pipeline mode is more complex because it involves managing a queue of pending queries and finding which result corresponds to which query in the queue.

Known Issues & Resolutions

Default Hash Partitions - No supported

Postgres doesn't support default hash partition

Default Hash Partition

```
1 pspmondb.pspadm> CREATE TABLE pspadm.psp_pstub_pay_item_def PARTITION OF pspadm.psp_pstub_pay_item DEFAULT
2 [2023-03-11 00:04:34] [42P16] ERROR: a hash-partitioned table may not have a default partition
```

Null Partition Key

Records with null partition results in not-null constraints violation

Null Partition Key

```
1 pspmondb.pspadm> INSERT INTO PSPADM.PSP_PSTUB_PAY_ITEM (PSTUB_PAY_ITEM_SEQ, VERSION, CREATOR_ID, CREATED_DATE,
2 MODIFIER_ID,
3                                     MODIFIED_DATE, REALM_ID, ACCT_NAME, EMPLOYEE_PAID,
4 INCOME_SUBJECT_TO_TAX, NAME,
5                                     PAYROLL_ITEM_LIST_ID, RATE, TYPE, WAGE_BASE, Y_T_D,
6 PAYSTUB_FK, CUR_AMT, QTY_AMT,
7                                     QTY_TIME, COMPANY_FK)
8     VALUES ('6b85e383-b1c2-4319-a195-89ac307f8c6e', 0, 'QBDTAdapter',
9             TO_TIMESTAMP('2013-08-07 15:19:12.948000', 'YYYY-MM-DD HH24:MI:SS.FF6'), 'QBDTAdapter',
10            TO_TIMESTAMP('2013-08-07 15:19:12.948000', 'YYYY-MM-DD HH24:MI:SS.FF6'), -1, null, 0,
11            0.0000,
12            'Social Security Employee', '196608-1106756535', null, 'Tax', 0.0000, -1072.6000,
13            '16ce581c-0ca0-40aa-9180-47ad21d651e0', -73.9600, null, null, null)
14 [2023-03-11 00:04:12] [23502] ERROR: null value in column "company_fk" of relation "psp_pstub_pay_item_p0"
15 violates not-null constraint
16 [2023-03-11 00:04:12] Detail: Failing row contains (6b85e383-b1c2-4319-a195-89ac307f8c6e, 0, QBDTAdapter, 2013-
17 08-07 15:19:12.948, QBDTAdapter, 2013-08-07 15:19:12.948, -1, null, 0, 0.0000, Social Security Employee, 196608-
```

```
1106756535, null, Tax, 0.0000, -1072.6000, 16ce581c-0ca0-40aa-9180-47ad21d651e0, -73.9600, null, null, null).
```

Solution

Workaround

DMS replication task mapping rules has been enhanced to filter records with null company_fk

Replication Task - Mapping Rules - Without Filter	Replication Task - Mapping Rules - With Filter
Replication Task - Mapping Rules - Without Filter <pre>1 { 2 "rule-type": "selection", 3 "rule-id": "644782231", 4 "rule-name": "644782231", 5 "object-locator": 6 { 7 "schema-name": "PSPADM", 8 "table-name": "PSP_PSTUB_PAY_ITEM" 9 }, 10 "rule-action": "include", 11 "filters": [] 12 }</pre>	Replication Task - Mapping Rules with Filter <pre>1 { 2 "rule-type": "selection", 3 "rule-id": "644782231", 4 "rule-name": "644782231", 5 "object-locator": 6 { 7 "schema-name": "PSPADM", 8 "table-name": "PSP_PSTUB_PAY_ITEM" 9 }, 10 "rule-action": "include", 11 "filters": 12 [13 { 14 "filter-type": "source", 15 "column-name": "COMPANY_FK", 16 "filter-conditions": 17 [18 { 19 "filter-operator": "notnull" 20 } 21] 22 } 23] 24 }</pre>

Long Term

Fix all the records with NULL partition key (company_fk) in the source

Duplicate key violates unique constraints

Duplicate Key Issue

```
1 2023-03-07 04:15:24 UTC:10.77.34.36(41252):postgres@pspmongodb:[29841]:STATEMENT: COPY "pspadm"."psp_compensation"
2 FROM STDIN WITH DELIMITER ',' CSV NULL 'attNULL' ESCAPE '\
3
4 2023-03-07 04:15:25 UTC:10.77.34.36(41326):postgres@pspmongodb:[29851]:ERROR:  duplicate key value violates unique
  constraint "psp_compensation_p5_pkey"
5
6 2023-03-07 04:15:25 UTC:10.77.34.36(41326):postgres@pspmongodb:[29851]:DETAIL:  Key (company_fk, compensation_seq)=(
7 515b07db-2986-4de7-8c2b-3a1fdd0f5bcd, 94e7286d-73e2-4ee9-929d-35eb91fd94aa) already exists.
```

Solution

- Ensure parallel tasks are not copying the same data sets
- Ensure the target table is empty

CLOB column value truncated

Clob column truncated

```
1 1678834453000,2023-03-14T22:54:13 [SOURCE_UNLOAD ]W: Oracle error code is '1406' ORA-01406: fetched column
  value was truncated (oracle_endpoint_unload.c:235)
2 1678834453000,2023-03-14T22:54:13 [SOURCE_UNLOAD ]E: ORA-01406: fetched column value was truncated [1022308]
  (oracle_endpoint_unload.c:235)
```

3 1678834453000,2023-03-14T22:54:13 [SOURCE_UNLOAD]W: The value of column 'MESSAGE' in table 'PSPADM.PSP_ENTITLEMENT_MESSAGE' was truncated to length 32768 (oracle_endpoint_utils.c:2838)

Solution

Include LOB columns in replication [Info](#)

- ☐ Don't include LOB columns
- ☐ Full LOB mode
- ☒ Limited LOB mode

Maximum LOB size (KB) [Info](#)

32

☐ Turn on validation

Choose this setting if you want AWS DMS to compare the data at the source and the target immediately after it performs a full data load. Validation ensures that your data was migrated accurately, but it requires additional time to complete.

- Change the replication settings to "Full LOB Mode" from "Limited LOB Mode" (Yet to tested)

RDS Trigger Error

RDS Trigger Error

```
1 1678617652000,2023-03-12T10:40:52 [SOURCE_CAPTURE ]I: Set corrupted redo log data block error as fatal error
  (oracle_endpoint_imp.c:1041)
2 1678617652000,2023-03-12T10:40:52 [SOURCE_CAPTURE ]W: Oracle error code is '4088' ORA-04088: error during
  execution of trigger 'RDSADMIN.RDS_DDL_TRIGGER2' ORA-00604: error occurred at recursive SQL level 1 ORA-20900:
  Invalid path used for directory: awsdms_dir_test ORA-06512: at ""RDSADMIN.RDSADMIN_TRIGGER_UTIL"", line 590 ORA-
  06512: at line 2 (oracle_endpoint_utils.c:3207)"
3 1678617652000,2023-03-12T10:40:52 [SOURCE_CAPTURE ]E: ORA-04088: error during execution of trigger
  'RDSADMIN.RDS_DDL_TRIGGER2' ORA-00604: error occurred at recursive SQL level 1 ORA-20900: Invalid path used for
  directory: awsdms_dir_test ORA-06512: at ""RDSADMIN.RDSADMIN_TRIGGER_UTIL"", line 590 ORA-06512: at line 2
  [1022307] (oracle_endpoint_utils.c:3207)"
4 1678617655000,2023-03-12T10:40:55 [TASK_MANAGER ]I: Postgres apply or data error. Failing stream component to
  retry at subtask 0, component st_0_Z3U2AAR45AE5R44HW6CILCLT7MKNJPSNEJXY7NI [1022521] (subtask.c:1409)"
5 1678617655000,2023-03-12T10:40:55 [TARGET_APPLY ]I: Target component
  st_0_Z3U2AAR45AE5R44HW6CILCLT7MKNJPSNEJXY7NI was detached because of postgres data or apply error. Will try to
  reattach (subtask.c:1535)
6 1678617661000,2023-03-12T10:41:01 [TARGET_APPLY ]I: Error_State: During startup error state is set to '1' and
  expiration time is already set. (odbc_endpoint_imp.c:5950)
7 1678617959000,2023-03-12T10:45:59 [TASK_MANAGER ]I: Postgres apply or data error. Failing stream component to
  retry at subtask 0, component st_0_Z3U2AAR45AE5R44HW6CILCLT7MKNJPSNEJXY7NI [1022521] (subtask.c:1409)"
8 1678617959000,2023-03-12T10:45:59 [TARGET_APPLY ]I: Target component
  st_0_Z3U2AAR45AE5R44HW6CILCLT7MKNJPSNEJXY7NI was detached because of postgres data or apply error. Will try to
  reattach (subtask.c:1535)
```

Solution

In Progress

Incorrect Extra Connection Attribute

Endpoint settings (JSON)

Use endpoint settings to specify additional connection attributes. Extra connection attribute (ECA) will be deprecated in the future, replaced by endpoint settings.

```
{
  "ExtraArchivedLogDestId": [],
  "DatabaseName": "PSPUEP03",
  "Port": 1521,
  "ServerName": "pspuwp03.cerpnqmbpqfla.us-east-2.rds.amazonaws.com",
  "Username": "intuadmin"
}
```

[Info](#)

Use extra connection attributes to define additional endpoint settings for AWS DMS

```
useLogMinerReader=true&useFile=Y&replacePathPrefix=true&usePathPrefix=/rdsdbdata/db/PSPUEP03_A/&useAlternateFolderForOnline=true&oraclePat
hPrefix=/rdsdbdata/db/PSPUEP03_A/&accessAlternateDirectory=false
```

Solution

AWS DMS uses LogMiner by default for change data capture (CDC) task. But PSP uses [Binary Reader](#) for below reasons

- Lower impact on source system I/O and CPU
- Better CDC performance

[Recommended Oracle Extra Connection Attributes](#)

For RDS Oracle versions 11.2 and 12.1	For RDS Oracle versions 12.2, 18.0, and 19.0
ECA <pre>1 useLogminerReader=N;useBfile=Y;accessAlternateDirectly=false;useAlternateFolderForOnline=true; 2 oraclePathPrefix=/rdsdbdata/db/{\${DATABASE_NAME}}_A/;usePathPrefix=/rdsdbdata/log/;replacePathPrefix=true;</pre>	ECA <pre>1 useLogminerReader=N;useBfile=Y</pre>

Replication Instance:

- maintenance window we can only disable 'automatic version updates' , **automatic version updates** set to 'off'.
- We can't disable weekly maintenance. If any maintenance activities are pending DMS instance will shutdown and restart.

Postgres to Oracle Migration:

- DB parameter **wal_sender_timeout** to be considered for Postgres to Oracle replication in addition to max_worker_processes.

wal_sender_timeout=0,

max_slot_wal_keep_size should be default

DMS Replication Parameters :

Maximum number of tables to load in parallel : 8,

Commit rate during full load : 10000

MemoryLimitTotal : 2048,

MemoryKeepTime : 120.

MemoryLimitTotal – Sets the maximum size (in MB) that all transactions can occupy in memory before being written to disk. The default value is 1024.

MemoryKeepTime – Sets the maximum time in seconds that each transaction can stay in memory before being written to disk. The duration is calculated from the time that AWS DMS started capturing the transaction. The default value is 60.